Sampling Completeness in Generative Search: Brand and Cited-Domain Accumulation under Repeated Queries

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Abstract

Under repeated identical buying questions, brand accumulation approaches a plateau while cited-domain accumulation keeps rising: in four deep prompt–model cells, ten runs reveal 91–100% of the brands observed across twenty-four, while cited-domain curves are still rising at the twenty-four-run endpoint, where a single run shows 16–23% of the observed domains; at breadth, one run shows 80% of the five-run brand set.

1 Accumulation in repeated answers

Auditors, brand trackers, and researchers sample LLM answers to commercial questions and report what they see. Every such report carries an implicit sampling claim about how much of what the model would say the collected runs have seen. Prior work established that identical prompts can yield different answers run to run [1, 5], and that repetition inside a single prompt changes model behaviour [2]. What has not been measured is the accumulation structure of repeated answers: how fast the set of distinct brands, and the set of distinct cited domains, grows as runs accumulate.

We treat runs as sampling units and brands (or cited domains) as species, and apply the classical accumulation toolkit [3, 4]: the exact rarefaction curve $A(k)$, the expected number of distinct items after k of the n collected runs, averaged over run subsets; the final rarefaction increment $A(n) - A(n-1) = Q_1/n$, where Q_1 counts items seen in exactly one run; and the Chao2 richness estimate, $S_{\text{obs}} + \frac{n-1}{n} Q_1^2/(2Q_2)$, with the bias-corrected form $S_{\text{obs}} + \frac{n-1}{n} Q_1(Q_1 - 1)/2$ when $Q_2 = 0$. Chao2 is a lower-bound estimator under its framework; rarefaction interpolates within the collected sample and predicts nothing about an unobserved next run. In particular, $Q_1 = 0$ means every observed item occurred in at least two runs; it does not certify that no unobserved item exists. All estimators are closed-form.

Figure 1 carries the result. Within the collected samples, brand curves approach a plateau: the final-run rarefaction increment is zero in all four deep cells. Cited-domain curves are still rising at the same endpoint, with 59–84% of the Chao2 estimate observed.

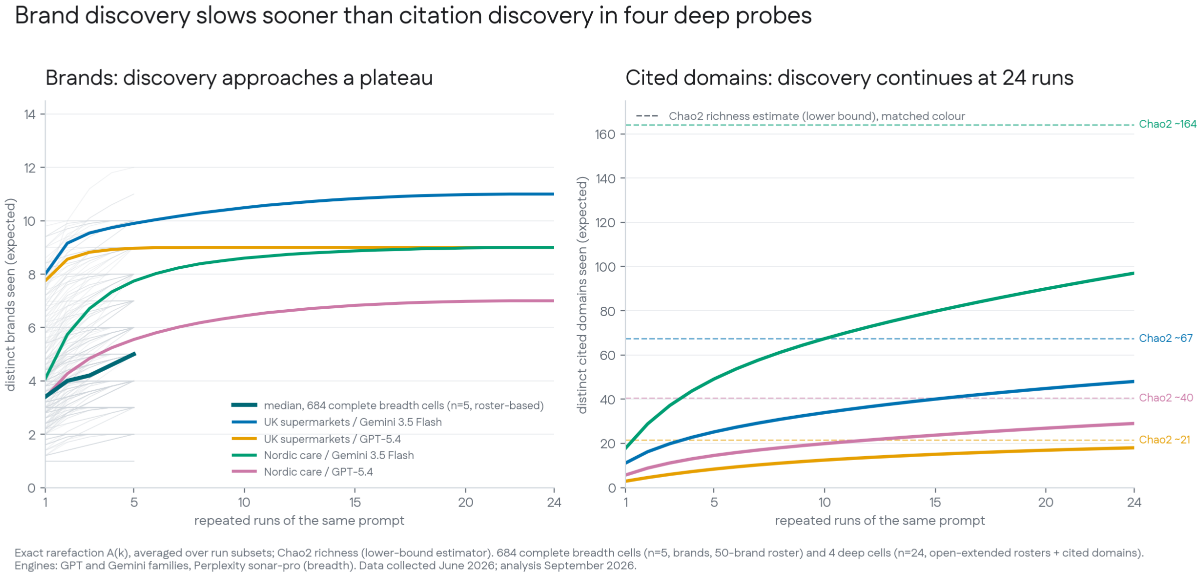

Figure 1: Expected accumulation of distinct brands (left) and distinct cited domains (right) under repeated identical prompts. The deep comparison covers two prompts on two models, 24 runs per combination, with each prompt–model cell drawn in its own colour, consistent across panels. By ten runs, expected brand discovery reached 91–100% of each observed 24-run brand set (open-extended rosters, Section 2); cited-domain curves continued rising at run 24. Dashed lines show Chao2 richness estimates, lower bounds under that framework, never known totals. Rarefaction averages over run subsets and does not establish that further runs would reveal nothing new. The grey background curves are the 705 retained breadth cells ($n \leq 5$, roster-based) with their median in dark teal; they are supporting material for the deep comparison. Data collected June 2026; analysis September 2026.

2 Data and results

Breadth: brand curves in 705 retained cells. We reanalyse our public Category Ownership deposit [7]: 250 brand-free category buying questions ("What is the best CRM for small businesses?") across five industries, put to GPT-5.2, Gemini 3 Flash, and Perplexity sonar-pro five times each in June 2026, with brand mentions extracted by the deposit's own 50-brand alias roster (3,750 responses). Of 750 planned question $\times$ model cells, 737 hold at least one response, 705 hold four or more runs, and 684 hold all five; each statistic below names its denominator. The breadth analysis measures accumulation among the deposit's 50 monitored brands: brands outside this extraction roster are not counted, so these curves do not estimate the model's unrestricted brand repertoire. They describe the object a fixed-roster tracker observes. Three counting results:

- The monitored set is quickly seen. The median cell names 5 distinct roster brands across five runs, and the median observed-to-Chao2 ratio at $n = 5$ is 100% (IQR 95–100%). Because $Q_1 = 0$ makes the Chao2 correction vanish mechanically, this ratio is a consistency statement, never proof of exhaustion. A single run shows 80% of the five-run set (median $A(1)/A(5)$ over the 684 complete cells).
- The tail is thin but real. In 332 of the 684 complete cells (49%; 339 of all 705 retained, 48%),

at least one brand occurred in only one observed run ($Q_1 > 0$): openai 56%, perplexity 47%, gemini 43% of cells.

- One model is not the market. For one query in four, a single model's five-run repertoire covers at most 83% of the three-model union (237 queries with all three models).

A case-insensitive dedup of brand strings changes none of these numbers.

Depth: 4 cells at $n = 24$, open-extended extraction. Two recommendation prompts (UK supermarkets; Nordic care providers), each put 24 times to GPT-5.4 and Gemini 3.5 Flash with web search, June 2026. The probes' stored extraction matched fixed rosters (10 and 6 names), and the observed plateaus sat one brand under those ceilings, so we then audited the extraction: open mining of capitalized candidate strings from the raw responses, with genuine providers adjudicated by reading their contexts. The audit recovered six organizations outside the rosters and one roster brand (Iceland) that the stored extraction had missed. Table 1 reconciles every cell; aliases collapse operating brands to parents (Vardaga and Nytida count as Ambea; M&S covers Marks & Spencer spellings).

Table 1: Extraction reconciliation per deep cell.

Cell	Stored	Corrected	Entities added
UK / Gemini 3.5 Flash	9	11	M&S; Iceland (roster brand, missed)
UK / GPT-5.4	8	9	M&S
Nordic / Gemini 3.5 Flash	5	9	Ersta Diakoni, Blomsterfonden, Stora Sköndal, Bräcke Diakoni
Nordic / GPT-5.4	5	7	Ersta Diakoni, Aleris

With the extended rosters the verdict is unchanged in direction and sharper in number: all four cells reach $Q_1 = 0$, and expected brand discovery is at 91–100% of each observed 24-run set by run ten. Domains, which the probes recorded openly from actual citations, carry no roster ceiling: the same four cells have Q_1 of 18, 6, 41, and 12 at run 24; observed surfaces of 48, 18, 97, and 29 domains against Chao2 estimates of 67, 21, 164, and 40 (71%, 84%, 59%, 72% observed-to-estimate); a single run shows 16–23% of the 24-run surface, and the within-sample final increment Q_1/n is still between 0.25 and 1.7 new domains per run at the endpoint.

The contrast is the finding, stated within its window: across these four cells, brand accumulation approached its plateau well before the 24-run endpoint, and cited-domain accumulation did not. The breadth corpus stores no citations, so the matched brand–domain contrast rests on the four deep cells; the 80% and 16–23% single-run figures also refer to different horizons (five and twenty-four runs) and are descriptive, never a controlled same-horizon comparison.

3 Related work

Non-determinism of repeated LLM queries is established [1], and our earlier work decomposes its variance and derives iteration requirements for stable means [5, 6]. This note measures set growth; the earlier work treats mean stability, and the two require different sampling budgets. Leviathan et al. [2] show repetition inside one prompt improves accuracy; we measure repetition across calls. Citation instability in generative search is documented on single runs [8, 9]; the accumulation view quantifies how much of the citation record any fixed sampling budget has observed. The estimators are the species-accumulation classics [3, 4].

4 Conclusion

In four prompt–model combinations, brand accumulation approached a plateau sooner than cited-domain accumulation. Following an extraction audit, ten runs revealed 91–100% of the brands observed across 24 runs, while cited-domain curves continued rising at the sampling endpoint. These results indicate that brand-set discovery and citation mapping can require different sampling budgets, which is directly actionable for anyone buying or selling repeated-query audits. They do not establish an exhaustive brand repertoire, and they do not determine the repeats needed to estimate recommendation frequencies precisely; set discovery and frequency estimation are different measurements needing different evidence [6]. Replication across more prompts, categories, languages, and models is needed to assess how broadly the observed contrast holds.

Future directions. (1) A breadth replication of the domain curves with citations recorded (the collection this note most wants; a six-engine replication at $n = 15$ with open extraction pre-registered is in progress). (2) Per-language curves on our 16-language repeated-query corpora. (3) Repertoire churn across model versions and time. (4) Accumulation-based stopping rules as audit budget policy, with their extrapolation assumptions stated. (5) Decoding-parameter effects on repertoire size. (6) In-the-wild replication on production audit data. (7) Whether prompt repetition in the sense of [2] changes the repertoire.

Declarations

Competing interests. The author is CEO of Rankfor.AI, which sells AI-visibility measurement, including repeated-query brand audits; this note bears on how such measurement should be priced and sampled. **Data and code.** The breadth corpus is our public deposit (Zenodo, 10.5281/zenodo.20788142, CC BY 4.0); all analysis code, per-cell tables, the extraction audit, and this note are at `https://github.com/Rankfor/rankfor-open/tree/main/research/recommendation-saturation`. **Provenance.** Secondary analysis of the author's own deposits; the accumulation analysis appears in no prior manuscript.

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